

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of (B. Sc) Examination, May, 2018

Course code & Name BE 402 & Introductory Developmental Biology

Date: 05/05/2018 Day: Saturday Time: 10.00 AM to 10.20 AM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		10
1.	Capacited mammalian sperm must penetrate the _____ and bind to the Zona pellucida before undergoing the acrosome reaction.		01
	(i) Cumulus	(ii) Egg cell membrane	
	(iii) Vitelline envelope	(iv) Extracellular matrix	
2.	_____ are cell adhesion proteins that some cells use for movement during their development.		01
	(i) Glycospheres	(ii) Cadherins	
	(iii) Receptors	(iv) Phospholipids	
3.	What type of cleavage is used in frog embryos?		01
	(i) radial	(ii) rotational	
	(iii) spiral	(iv) superficial	
4.	The darkly pigmented end of a <i>Xenopus</i> egg is called		01
	(i) dorsal	(ii) yolk	
	(iii) the vegetal pole	(iv) animal pole	

5.	The maternal dorsalizing factors act by stimulating a pathway that leads to the local stabilization of Which protein in the future dorsal region?		01
	(i) Vg-1	(ii) β -catenin	
	(iii) BMP4	(iv) Xwnt-8	
6.	How many cleavages are completed in 16 cell stages of frog's egg ?		01
	(i) 4	(ii) 3	
	(iii) 5	(iv) 8	
7.	Assuming the gamete amount of DNA in the human to be 2.0 picograms, how many picograms of DNA is there in the nucleus of an oocyte after ovulation but prior to fertilization?		01
	(i) 1.0	(ii) 2.0	
	(iii) 6.0	(iv) 4.0	
8.	When an immature cell matures into a specialized cell line, this is called		01
	(i) metamorphosis.	(ii) induction.	
	(iii) differentiation.	(iv) metaplasia	
9.	Which of the following tissues can be derived from mesoderm?		01
	(i) a carbohydrate	(ii) Blood cells	
	(iii) Fibrous tissue	(iv) Simple squamous epithelium	
10.	The biological reason underlying meiosis in the human organism is		01
	(i) to make reproduction much more complicated	(ii) because it makes reproduction fun.	
	(iii) to prevent trisomy of genes	(iv) to increase diversity and variability	

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IV Semester of (B. Sc.) Examination, May 2018

Course code & Name BE 402 & Introductory Developmental Biology

Date: 05/05/2018 Day: Saturday Time: 10.21 AM To 12.00 PM Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q – II Answer the following questions as directed (attempt any 5)		10
1.	What are major steps for embryonic development? Which 3 signaling pathways are important for induction and formation of the organizer?	02
2.	What are the advantages and disadvantages of using drosophila as a model system in Developmental Biology?	02
3.	What are the basic differences between Epigenesis and Preformation?	02
4.	What is the fast block to polyspermy, and what is the molecular basis of it?	02
5.	What is the acrosomal reaction, and what are its consequences?	02
6.	What are the differences between totipotent and multipotent stem cells?	02
7.	What is cloning? What are the Ethical Issues regarding Human Cloning?	02
SECTION – II		
Q-III Answer the following questions as directed (attempt any 5)		15
1.	After the blastula stage what is the next stage? What is the morphological feature that defines this stage?	03
2.	Describe the Spemann-Mangold experiment. What key aspect of the experiment showed that induction was occurring, i.e. that the transplant was changing the normal cell fates of the surrounding tissues?	03
3.	What are some of the major structures of a mature sperm?	03
4.	How can stem cells be used to treat diseases and injuries? Why is there so much controversy around embryonic stem cell research?	03

5.	What are the major questions of Developmental Biology? What are advantages and disadvantages of using mouse as a model system?	03
6.	Write a short note on Sugar alcohol Somatic cell nuclear transfer (SCNT).	03
7.	Write a short note on Model organisms:	03